Hotspot Analysis

Andrea Cattarinich S5137057

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1 Hotspot Analysis

The goal of this analysis is to identify the parts of the program that consume the most execution time. The measurements were performed using two methods:

- **Manually**: using `omp_get_wtime()`
- **Profiler**: using `Intel VTune`

Either two methods have been used with with [$N = 10000$] on [WSL2-workstation] and produced [similar results].

Table 1: Manually

Function	Time (s)	Percentage (%)
fillInput	0.00185	0.026
setOutputZero1	0.00012	0.002
setOutputZero2	0.00011	0.002
DFT	3.62358	50.213
IDFT	3.59071	49.758
checkResults	0.00002	0.000
Total	7.21639	100.000

Table 2: Intel VTune Profiler

Function	Time (s)	Percentage (%)
fillInput	0.xxx	0.026
setOutputZero1	0.xxx	0.002
setOutputZero2	0.xxx	0.002
DFT	xx.xxx	50.213
IDFT	xx.xxx	49.758
checkResults	0.xxx	0.000
Total	xxx.xxx	100.000

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1.1 Conclusion

The main hotspot is the DFT() function and its inverse IDFT(), which together account for over 99% of the total execution time. All other functions have negligible impact. This indicates that **parallelization and vectorization efforts should focus on the double loop inside DFT()**.